NAM HYUN KANG

SOFTWARE ENGINEER | EMBEDDED & DEVICE SOFTWARE

Yongin-si, South Korea | rkdskagus123@gmail.com

PROFESSIONAL SUMMARY

Software engineer with 5+ years of software development experience, including embedded firmware and Linux/Android device software. Work includes C++ sensor control, Java-based device services, equipment monitoring, and scenario-based testing with QA handoff.

TECHNICAL SKILLS

Programming: C, C++, Java, Python, JavaScript, C#

Operating systems: Linux (Ubuntu), Android, RTOS

Interfaces & frameworks: I2C, TR-069, REST APIs, RTSP, WebRTC, FastAPI

Development & testing: Git, code review, manual testing, oscilloscope

Tools used in projects: MediaMTX, Visual Studio, Eclipse, Android Studio

PROFESSIONAL EXPERIENCE

CanTops | Software Engineer Aug 2024 - Present

- Developed C++ firmware for ToF sensor evaluation and display control over I2C on a company-developed real-time operating system (RTOS). Handled sensor bring-up and module integration, and contributed to vibration and tilt sensor integration for overhead hoist transport (OHT) monitoring.
- Investigated burst-read failures during a 4x4 ToF sensor evaluation. Oscilloscope measurements and code analysis identified an OS initialization routine that overwrote the I2C speed setting. Patched the routine, added diagnostic logs, and verified normal data reception. Built a GUI to parse and plot the readings and pushed the changes to the Git repository. The ToF option was not selected after a change in product direction.
- Reimplemented Windows Forms monitoring and configuration functions as a web application for OHT- and rail-mounted data-acquisition devices. Integrated live video and sensor views, historical data, and device settings; used MediaMTX to relay RTSP video to browsers over WebRTC.
- Fixed JavaScript incompatibilities on customer Linux PCs running older Firefox versions within one week. Verified video playback, sensor views, historical records, and device settings in the customer environment.
- Developed selected firmware components for wireless LAN equipment. Built an embedded web server and web UI using JavaScript, Python, and FastAPI for device settings, status and log viewing, and FTP/CLI control.
- Built a C# sensor diagnostic tool for live visualization and raw-data analysis. Separated reception, parsing, and UI updates into dedicated threads, using a circular buffer and queue.

PROFESSIONAL EXPERIENCE — CONTINUED

Innopia Technologies, Inc. | Software Engineer Apr 2021 - Sep 2023

- Proposed and implemented a redesign of a legacy set-top-box library whose hard-coded structure made feature changes difficult. The revised software was incorporated into a shipped product.
- Reimplemented existing Java code using object-oriented design to meet updated Android requirements. Revised communication between in-house applications and the software structure to support future protocol extensions.
- Developed and extended TR-069 services and device-management data models for set-top boxes using C++ and Java on Linux/Android.

Fourth Logic Incorporated | Software Engineer May 2020 - Sep 2020

- Implemented and optimized C++ geometry routines for precision measurement in an AI image-processing platform, including polygon intersections and volume calculations involving spheres.

Unichal Inc. | Software Engineer May 2018 - Jan 2019

- Developed firmware for company projects.

DEVELOPMENT & VALIDATION

Created and manually ran test scenarios covering normal operation and deviations from the expected workflow, then shared the scenarios with QA for further testing. Used Git for version control and submitted code changes for review.

EDUCATION & ADDITIONAL INFORMATION

Dankook University | Bachelor's degree, Mobile Systems Engineering | Mar 2013 - Feb 2018

Simon Fraser University | Mechanical Engineering studies; no degree awarded | 2009 - 2012

Engineer Information Processing (Korea) | May 2017 | Korean: Native | English: Native

Career break: Prepared to start a business between Innopia Technologies and CanTops.